

Silicone Elastomer with Surface-Enriched, Nonleaching Amphiphilic Side Chains for Inhibiting Marine Biofouling

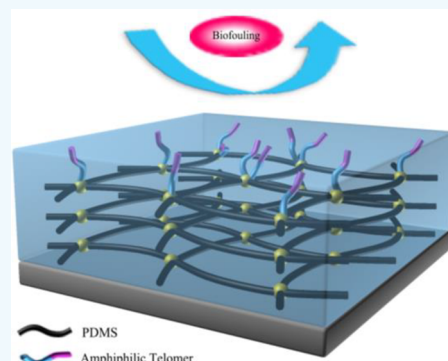
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Supporting Information

ABSTRACT: Silicone elastomer is one of the ecofriendly fouling release materials. However, it suffers from poor fouling resistant performance during idle periods. We developed a silicone elastomer with self-stratifying, nonleaching amphiphilic side chains by grafting telomer of dodecafluoroheptyl methacrylate (DFMA), poly(ethylene glycol) methyl ether methacrylate (PEGMA), and 3-mercaptopropyl trimethoxysilane (KH590) to bis-silanol terminated silicone. The amphiphilic telomer can be self-enriched on the surface during coating formation because the fluorocarbon segment with low surface energy is incompatible with silicone. Meanwhile, the telomer with KH590 can cross-link to silicone so that it is nonleaching. Such modified silicone elastomer has a low surface energy and low elastic modulus close to the unmodified one because most of the amphiphilic telomers are on the surface, so the former as a coating still has excellent fouling release performance. Moreover, it has remarkable fouling resistance toward marine bacterial biofilm and diatoms under a suitable molar ratio of DFMA and PEGMA (GF1P2). Such a modified silicone elastomer is expected to find application to inhibit marine biofouling.

KEYWORDS: silicone elastomer, amphiphiles, telomerization, marine biofouling, fouling release



1. INTRODUCTION

Marine biofouling is a serious problem in marine industries and activities. It significantly increases hydrodynamic drag, fuel consumption, and greenhouse gas emission of ships.¹ So far, using an antifouling coating is the most convenient and effective way to combat biofouling.^{2,3} Since the global ban of highly toxic tin-based coatings,⁴ much effort has been paid to develop ecofriendly antifouling materials such as amphiphilic copolymers,^{5,6} protein resistance polymers,^{7,8} antibacterial polymers,^{9,10} and degradable polymers.^{11,12} Particularly, the silicone elastomer-based fouling release coatings have gained much attention in that they are ecofriendly materials with drag reducing ability.^{13,14}

The fouling release ability of silicone elastomer is mainly attributed to its low surface energy and low elastic modulus,^{15–18} which weakens the adhesion strength of marine organisms, allowing for easy removal. However, the removal of accumulated fouling organisms needs high shear stress, so silicone elastomers have poor fouling release performance under static conditions. Moreover, silicone elastomers cannot inhibit the accumulation of the slime layer, consisting of diatoms and bacteria, which is hard to detach even at the high speed of ships.^{19–21} Therefore, it is essential to improve the fouling resistance of silicone elastomers.

Amphiphilic polymers possessing both hydrophobic and hydrophilic components are effective in inhibiting fouling organisms.^{22–24} Thus, combining amphiphilic polymers with polydimethylsiloxane (PDMS) is an effective strategy. Kiil et

al.²⁵ physically mixed the polyethylene glycol (PEG)-based amphiphiles in PDMS coatings and investigated the diffusion coefficient of amphiphiles through coating films. Their study demonstrated that the fouling resistant performance largely depended on the hydrophobic block of the amphiphilic compound. However, fouling resistant efficiency decreased with such physical mixing because of the amphiphile's leaching. Grunlan et al.^{26–28} prepared a series of PEG-silane amphiphiles with cross-linking end groups and chemically grafted them into PDMS. The resultant coatings show excellent resistance to protein and biofilm adhesion. Because the amphiphiles contain oligodimethylsiloxane, which is compatible with the PDMS, a large proportion of them may be trapped in the bulk instead of being enriched on the surface. It is also reported that PEG and fluorocarbon are separately grafted into the PDMS network.^{29,30} In this case, it is difficult for PEG to migrate onto the surface because it has surface energy higher than that of PDMS, leading to poor fouling resistant performance.

In this present study, we synthesized an amphiphilic oligomer of poly(ethylene glycol) methyl ether methacrylate (PEGMA), dodecafluoroheptyl methacrylate (DFMA), and 3-mercaptopropyl trimethoxysilane (KH590) by telomerization, where the PEG segment, fluorocarbon segment, and trimethoxysilane moiety provide the telomer with fouling

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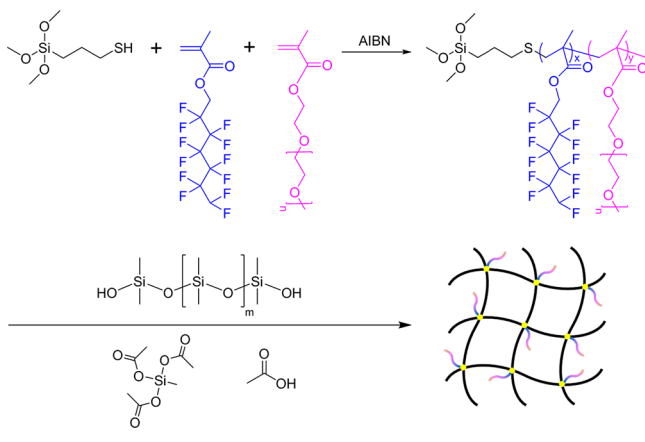
resistant ability, low surface energy, and cross-linking ability, respectively. The telomer was grafted to bis-silanol PDMS via condensation reaction. Before the coating is completely dry, the fluorocarbon segments with low surface energy and incompatibility with silicone drive the telomer to migrate onto the surface, yielding an amphiphile-enriched surface. Moreover, the cross-linking makes its nonleaching possible. Such a hierarchical structure is expected to exert fouling resistance of the nonleaching amphiphile and retain the fouling release ability of PDMS. We systemically studied the properties of the coating in terms of surface wettability, modulus, and fouling resistant performances. By this study, we aim to develop novel silicone coating with durable fouling resistant and fouling release abilities even under static conditions.

2. EXPERIMENTAL SECTION

2.1. Materials. MED-2000 comprised of bis-silanol terminated PDMS, methyltriacetoxysilane, ethyltriacetoxysilane, silica, and acetic acid from Nusil Technology were used as received. DFMA was purchased from Harbin Xuejia Fluorin Silicon Chemical Co., Ltd. and used as received. PEGMA ($M_n = 300$ g/mol) and 3-mercaptopropyl trimethoxysilane (KH590, Aldrich) were used as received. Azobis-(isobutyronitrile) (AIBN, Aladdin) was purified by recrystallization from methanol. Fibrinogen (fraction I from human plasma, $M_w = 340$ kDa, pI 5.5) was from Merck Chemicals. Tetrahydrofuran (THF) from Sinopharm was stirred with CaH_2 and distilled prior to use. Artificial seawater (ASW) was prepared according to ASTM D1141-98 (2003).

2.2. Synthesis of Amphiphilic Telomers. The telomerization of KH590, PEGMA, and DFMA was conducted following a procedure elsewhere.^{31,32} Typically, KH590 (0.73 g), PEGMA (5.89 g), DFMA (4.00 g), and AIBN (0.02 g) were dissolved in THF (20.0 mL) and introduced in a Schlenk tube (100 mL) with a magnetic stirring bar. The solution was degassed by three freeze–evacuate–thaw pump cycles. The tube was then filled with argon, and the reaction proceeded at 70 °C for 24 h. The resulting solution was purified by precipitations in hexane, and solvent was removed under vacuum at 40 °C. The obtained telomers were designated as GFxPy (x and y are the molar ratios of DFMA and PEGMA, respectively). The silyl ether end groups are on one end of the telomer (Scheme 1). Besides

Scheme 1. Synthetic Route of GFxPy and PDMS-GFxPy



GFxPy, telomer without PEGMA designated as GF and telomer without DFMA designated as GP were synthesized using a similar procedure. The molar composition and molecular weight of the telomers were determined by ^1H NMR spectra (Figure 1). The molecular weight is in the range of 2400–2800 g/mol. The corresponding data can be found in Table 1.

2.3. Preparation of PDMS-Based Coating with Grafted Telomer. The coating was prepared by condensation reaction

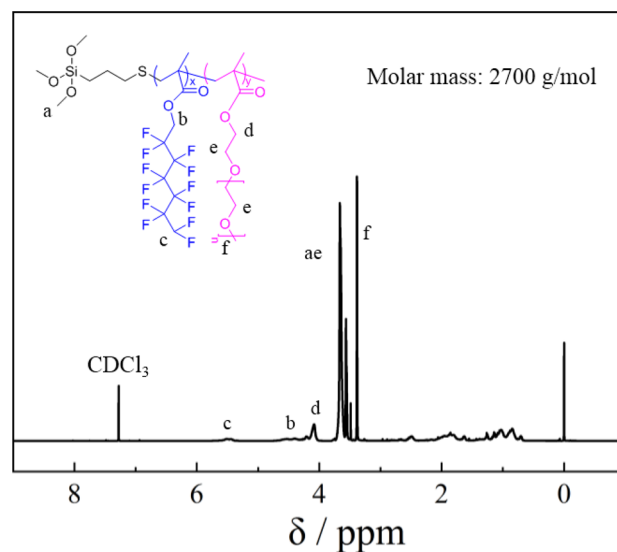


Figure 1. ^1H NMR spectra of GF1P2.

Table 1. Characterization Data of the Telomer

sample	DFMA/PEGMA/ KH590 ^a	DFMA/PEGMA/ KH590 ^b	molar mass ^c (g/mol)
GF2P1	5.3/2.7/1/	4.4/2.3/1	2600
GF1P1	4/4/1	3.5/3/1	2500
GF1P2	2.7/5.3/1/	2.4/5.2/1	2700
GF	8/0/1	6.5/0/1	2800
GP	0/8/1	0/7.5/1	2400

^aFeed molar ratio. ^bMolar ratio in telomers determined by ^1H NMR.

^cDetermined by ^1H NMR.

(Scheme 1). Typically, MED-2000 (10.00 g) and GFxPy (1.76 g) were dissolved in THF (100 mL) and stirred with a magnetic bar for 1 h. The solution was cast onto a silicon slide (for atomic force microscopy and fouling release test), 24-well polystyrene (PS) microplate (for bacterial biofilm test), and epoxy resin panel (for mass loss test) and spin-cast on AT-cut quartz crystal surface (for protein adsorption test), glass slide (for surface wettability), and silicon slide (for antibacterial assays and antidiatom tests). The coatings were cured at room temperature for 5 days. The coatings on silicon substrate and gold for QCM test have similar and enough thickness. Thus, the substrate has slight influence on the surface topology and surface chemistry.³³ The PDMS-based coating is designated as PDMS-GFxPy. Those with grafted GF or GP are designated as PDMS-GF or PDMS-GP. PDMS-GF/GP also contains 15 wt % of telomer (GF/GP).

3. CHARACTERIZATION

3.1. Proton Nuclear Magnetic Resonance Spectroscopy (^1H NMR). The chemical structure was characterized by ^1H NMR spectra on a Bruker AV600 NMR spectrometer with CDCl_3 as solvent.

3.2. X-ray Photoelectron Spectroscopy (XPS). XPS analysis was performed using an Axis Ultra DLD (Kratos Analytical) with a monochromate Al $K\alpha$ ($h\nu = 1486.6$ eV) source at takeoff angles of 90°. The X-ray gun was operated at the voltage of 15 kV and current of 5 mA. Binding energies were all referenced to the C 1s at 284.6 eV.

3.3. Contact Angle Measurements. Static water contact angle (WCA) measurements were performed with a Theta Auto 113 (KSV NIMA, Biolin) at room temperature by depositing a droplet of liquid (3 μL) on the sample surface using the sessile method. After immersion in ASW for different time, the contact angle was measured using the same method. An average contact angle was measured from three different regions of the same sample. Dynamic WCA was also performed according to a previous report.³⁴ Briefly, it was measured

by adding the volume of a water droplet from 1 to 4 μL and finally decreasing it. The advancing and receding WCAs were taken as the maximum and minimum stable angles, respectively. Advancing WCA is related to hydrophobicity, while receding WCA is related to hydrophilicity. Surface energy (SE) of PDMS coating was calculated from water and diiodomethane contact angle using the Owens–Wendt–Rabel–Kaelble method.³⁵

3.4. Atomic Force Microscopy (AFM) Measurements. The phase image of PDMS-GF α Py on the silicon slide was measured using an XE-100 AFM (Park Systems, Korea). The AFM phase image can be identified by the dark and light regions, where the dark or light region can reflect the phase separation of soft and hard segments. Noncontact mode was employed to obtain phase images on a scan size of $5.0 \times 5.0 \mu\text{m}$ (NCHR cantilever). Surface elastic modulus measurement was also conducted by AFM using Hertzian contact theory, which correlates the AFM force signal (F) to the indentation depth of the cantilever tip (δ) and elastic modulus (E) by the following equation:³⁶

$$F = \frac{E}{1 - \nu^2} \frac{\tan \alpha}{\sqrt{2}} \delta^2$$

where ν is the Poisson's ratio and is 0.5 for all coatings, α is the face angle of the four-sided pyramidal tip of the cantilever (22°), and δ is the indentation depth.

3.5. Fouling Release Test. The test was performed following ASTM D5618 (2011).³⁷ Five pseudobarnacles made of cylindrical aluminum studs (10 mm in diameter, 10 mm in height) were attached to the coating surface with two-component adhesive (Araldite Rapid Epoxy Adhesive, Huntsman). The adhesive was cured for 3 days under ambient conditions. The removal test was conducted at constant speed, and maximal shear force was obtained by using the force gauge (SUNDOO SH-500).

3.6. Quartz Crystal Microbalance with Dissipation (QCM-D). The QCM-D measurement of PDMS-GF α Py was performed using an E1 system (Q-sense AB, Sweden). The polymer solution in THF (1.0 wt %) was spin-cast on AT-cut quartz crystal surface and cured at room temperature for 5 days. ASW was used as a baseline, and 1 mg/mL protein solution was delivered to the surface. Briefly, the change of frequency (Δf) is correlated with the protein adsorption on quartz crystal, whereas the change in dissipation (ΔD) is integrated with the viscoelastic properties of the adsorbed protein layer.³⁸

3.7. Water-Induced Mass Loss. The mass loss test was conducted according to the previous reports.^{27,39} Generally, the dry PDMS-GF α Py coatings were weighed (W_0) before the coatings were immersed in ASW. After 2 weeks, the coatings were taken out and rinsed with deionized water. Afterward, they were dried under vacuum at room temperature for 2 days and then weighed (W_t). The mass loss was calculated by $(W_0 - W_t)/W_0 \times 100\%$. The dimensions of the cured coatings are $20 \times 20 \text{ mm}^2$, and the thickness is $\sim 300 \mu\text{m}$ for all coatings. Triplicate samples were used to calculate the average mass loss.

3.8. Antibacterial Assays. Marine *Pseudomonas* sp. was introduced to investigate the antibacterial performance of the coatings following our previous report.³⁴ The suspension was diluted to 10^8 cells/mL with ASW. PDMS-GF α Py coatings were immersed in bacterial suspension (1 mL). The bacteria were incubated at 30°C for 4 h. Subsequently, LIVE/DEAD BacLight Bacterial Viability Kit was used to stain the bacteria which settled on the coating surface. A fluorescence microscope (Scope A1, Zeiss) was used to examine the adhered bacteria. The number of bacteria was determined by three random fields of view on triplicate slides. Data were normalized to that of control sample.⁴⁰

3.9. Antibiofilm Test. Culture and incubation of the bacteria *Pseudomonas* sp. were conducted following our previous report.³⁴ Bacterial suspension at mid-log phase was diluted to 10^8 cells/mL with MB2216. Bacterial suspension (1 mL) was added to the sample well coated with PDMS-GF α Py coating on the bottom. Sealing films were applied, and plates were incubated in an incubator (Bluepard). The biofilm was allowed to incubate for 48 h at 30°C . The suspension was then removed, and the plates were gently rinsed with

ASW to remove any unattached biofilm. The biofilm was fixed with methanol and dried at room temperature for 1 h. Then, the biofilm was stained with crystal violet (0.5 mL) for 15 min, and the unstained crystal violet was washed (ASW). Then, images were taken using a digital camera. The percentage coverage was determined by ImageJ, and data were normalized to that of control samples. To quantify the biofilm grown on the coating, 33% of acetic acid (in water) was employed to extract the crystal violet in the adherent biofilm, and eluates were tested at $\lambda = 589 \text{ nm}$.

3.10. Antidiatom Test. The antidiatom property of the coating was evaluated using *Navicula incerta* (*N. incerta*). Coatings on glass slides were used to carry out antidiatom test. Guillard's F/2 culture medium was employed to cultivate *N. incerta* in plant growth chamber (Bluepard) at 23°C with a 12:12 h light-dark cycle. Glass slides coated by PDMS-GF α Py coating were placed in a 24-multiwell plate. One milliliter of diatom suspension (chlorophyll content: $0.50 \mu\text{g}/\text{mL}$) was added to each well and incubated for 24 h. The glass slides were gently washed with ASW to remove the unattached diatoms. Counting was performed using microscopy (Scope A1, Zeiss), and 10 random fields of view on triplicate slides were selected to quantify diatom density.

4. RESULTS AND DISCUSSION

Figure 1 shows the typical ^1H NMR spectra for the GF1P2 telomer. The peak at 3.59 ppm is attributed to the methoxy group on silane. The signal of fluorocarbon group can be observed at 4.37 and 5.47 ppm. Peaks at 4.08, 3.60, and 3.38 ppm can be assigned to the proton of CH_2 linked to the ester group and CH_2 and CH_3O of the PEG moiety, respectively. These results clearly indicate the successful synthesis of the telomer. Its molar mass was calculated to be $\sim 2700 \text{ g/mol}$. The telomers were subsequently cross-linked with bis-silanol terminated PDMS, generating a modified silicone coating (PDMS-GF α Py). To verify the reactivity of the telomer, the nonleaching property of modified silicone was first examined by the mass loss of the coating after immersion in ASW for 2 weeks (Figure S1). For PDMS, the mass loss was $\sim 0.5 \text{ wt } \%$, which could be attributed to the removal of some unreacted polysiloxane and residue solvent. In comparison, each PDMS-GF α Py has a mass loss (0.3–0.5 wt %) similar to that of nonerodible PDMS, indicating the telomer was grafted to the PDMS network and the modified PDMS coating exhibits nonleaching property.

To examine the surface composition of PDMS-GF α Py coatings, we conducted XPS spectra (Figure 2). The peaks at 150.5 and 100.4 eV are attributed Si (2s and 2p) in PDMS, and the peak at 685.7 eV is attributed to F (1s) in fluorocarbon groups. For PDMS and PDMS-GP, no F (1s) is detected because they do not contain any fluorocarbon. For PDMS-GF α Py, the peak of F (1s) can be observed, and the peak intensity of F (1s) increases with the content of fluoro groups. More importantly, all the PDMS-GF α Py exhibit F/Si values on the surface higher than those in the bulk (Table 2), indicating that the low surface energy of fluorocarbon serves as a driving force to migrate telomers to the surface. In this case, PEG moiety is also dragged upward to the surface because PEG and fluorocarbon segments are randomly distributed in the telomer chain. Note that because both the fluorocarbon and PEG have C–O peaks in XPS spectra, we cannot distinguish their contribution and thus cannot directly calculate the content of PEG on the surface.

We also studied the migration of the amphiphile onto the surface by measuring the WCA of PDMS-GF α Py coating with different immersion time in ASW. As shown in Figure S2, the static WCA of PDMS and PDMS-GF changed from $\sim 110^\circ$ to

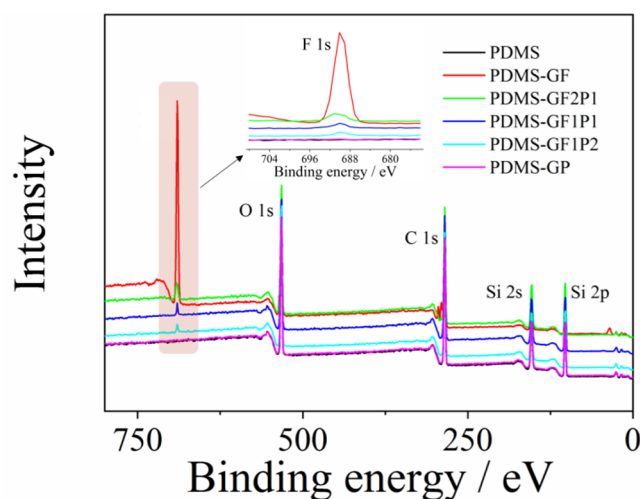


Figure 2. XPS scan spectra of PDMS-GF&Py coatings. The inset is F spectra at high resolution.

Table 2. Elemental Weight Ratio of F/Si on PDMS-GF&Py Coatings

sample	F/Si in bulk	F/Si on the surface
PDMS	0/100	0/100
PDMS-GF	16.2/100	141.2/100
PDMS-GF2P1	10.8/100	26.1/100
PDMS-GF1P1	7.5/100	17.4/100
PDMS-GF1P2	5.9/100	12.5/100
PDMS-GP	0/100	0/100

105° and ~115° to 110° due to the reconstruction of the surface.⁴¹ For PDMS-GP, it has a greater change from ~110° to 101°. However, the WCA of PDMS-GF&Py changes notably (>15°). Figure 3a shows the equilibrated static WCA. For

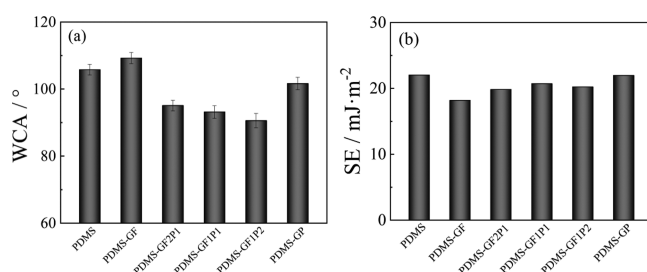


Figure 3. (a) Equilibrated static WCA of PDMS-GF&Py coatings after immersion in ASW and (b) SE of PDMS-GF&Py coatings.

PDMS, the WCA is ~105°, indicating the surface is hydrophobic. PDMS-GF exhibits a higher WCA of 110° because the DFMA group has strong tendency to migrate on the surface due to its low surface energy and incompatibility with silicone, which increases the hydrophobicity. PDMS-GP has a slight change of WCA (~4°) to that of PDMS even if the content of PEG telomer is ~15 wt %. This is probably because the PEG is buried in PDMS network. Interestingly, once the telomer contains both PEG and DFMA segments, they exhibit lower WCA, and the WCA decreases as PEG content increases (from GF2P1 to GF1P2), indicating the higher concentration of hydrophilic PEG moiety on the coating surface.

The dynamic WCA has a trend similar to that of the static WCA (Figure S3). The advancing WCA of PDMS decreases

from ~112° to 103° because of the reconstruction of Si-CH₃ on the surface. For PDMS-GF, its advancing WCA changes slightly because DFMA can enrich on the surface and increase the hydrophobicity. For coating containing both fluorocarbon and PEG moieties, the advancing WCA decreases as fluorocarbon decreases. On the other hand, the receding WCA of PDMS changes from ~60° to 47° while the receding WCA of PDMS-GF changes from ~75° to 60°. For PDMS-GP, it has a receding WCA close to that of PDMS. For coating containing both fluorocarbon and PEG moieties, the receding WCA decreases as PEG content increases. Note that the PEG content in PDMS-GF2P1, PDMS-GF1P1, and PDMS-GF1P2 is lower than that of PDMS-GP, clearly indicating the PEG is dragged upward to the surface by the linked DFMA group; that is, the coating is a self-stratifying system. We also examined the surface energy of the PDMS-GF&Py coatings (Figure 3b). PDMS has a surface energy of 22 mJ/m². PDMS-GP has a surface energy of 21 mJ/m², but PDMS-GF has a lower surface energy of 18 mJ/m² due to the migration of DFMA groups. For either PDMS-GF2P1 or PDMS-GF1P2, the surface energy is lower than 22 mJ/m². The result indicates that the chemically grafted amphiphilic telomer has a slight impact on surface energy of PDMS-GF&Py coatings, which is expected to facilitate the fouling release property.

To further verify the self-stratification effect of the PDMS-GF&Py coatings, AFM was used to study their phase images (Figure 4). For PDMS, an even phase was observed. In

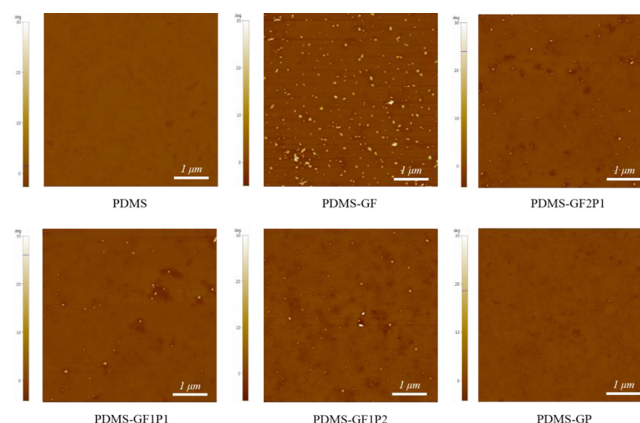


Figure 4. AFM phase image of PDMS-GF&Py coatings.

contrast, PDMS-GF exhibits large amounts of discontinuous light and dark regions (corresponding to the hard fluorocarbon segments and soft PDMS segments), indicating the phase segregation on the surface. As the content of fluorocarbon decreases, the light region becomes smaller because less fluorocarbon is on the surface. Anyhow, the current phase segregation data indicate the coating is a self-stratifying system where both the lower surface energy of fluorocarbon and incompatibility with PDMS matrix serve as the driving force for surface enrichment. Note that PDMS-GP has a similar phase image to PDMS. As reported before,⁴² PEG has surface energy higher than that of PDMS. Besides, the PEG in the current study has cross-linkable silane end groups, which can react with PDMS. Therefore, it is difficult for PEG to migrate onto the surface, and it is buried in the matrix in spite of its immiscibility with PDMS.

We also measured the surface elastic modulus (Figure 5 and Figure S4). PDMS has the lowest modulus of 0.9 MPa. PDMS-

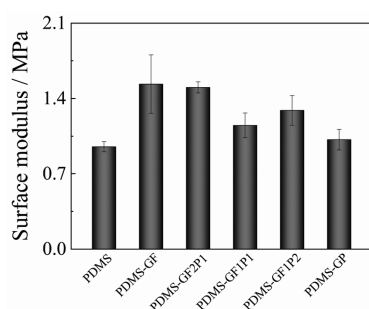


Figure 5. Surface modulus of PDMS-GF_xPy coatings.

GF has the highest modulus due to the rigid fluorocarbon groups on the surface. For PDMS-GP, its modulus is slightly higher than that of PDMS but lower than others. This is probably because the PEG chains are trapped in the bulk and have little influence on the surface modulus. PDMS-GF2P1, PDMS-GF1P1, and PDMS-GF1P2 have a higher modulus than that of PDMS due to the surface enrichment of the telomer. Further, the surface modulus of each sample is lower than 2.0 MPa, indicating the telomer does not essentially change the elastic and soft nature of silicone. With the low surface energy and low modulus, the coating may have good fouling release performance.

To verify whether the modified fouling resistant PDMS coatings have fouling release ability after the addition of telomers, the pseudobarnacle test was conducted (Figure 6). A

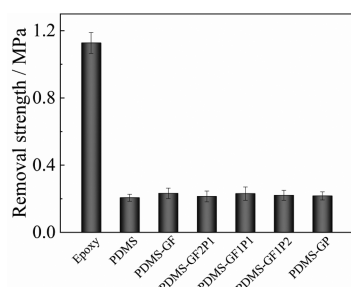


Figure 6. Removal strength of pseudobarnacles adhered to PDMS-GF_xPy coatings.

removal strength as large as 1.1 MPa is needed to remove pseudobarnacles from the epoxy substrate, while PDMS demonstrates a good fouling release performance with a low strength of ~ 0.2 MPa. For PDMS-GF_xPy, it exhibits removal strength slightly higher than that of PDMS. It is known that the fouling release ability is related to the surface energy and elastic modulus. As discussed above, the enrichment of telomers slightly increases the surface modulus so that the fouling release ability slightly decreases. PDMS-GF_xPy still exhibits low removal strength. In other words, the coating keeps the fouling release ability of PDMS. Actually, the adhesion strength of the coatings containing the telomer to substrate is similar to that of the traditional PDMS coating. Thus, they are adhesive to the ship body and can be considered in marine applications.

Because biofouling usually starts from protein adsorption and many fouling organisms secrete protein-based bioadhesive to settle on surfaces,⁵ we examined the protein resistance of the modified silicone elastomer by using QCM-D. Figure 7 shows the time dependences of frequency shift (Δf) and dissipation shift (ΔD) for the adsorption of fibrinogen on the coatings.

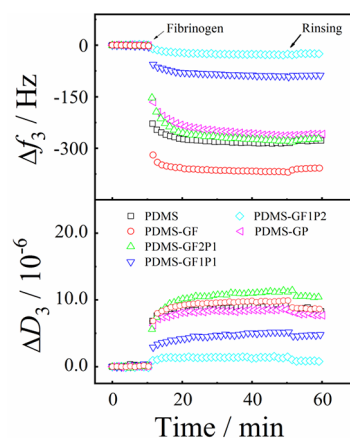


Figure 7. Change of frequency (Δf) and dissipation (ΔD) regarding adsorption of fibrinogen on PDMS-GF_xPy coatings.

Fibrinogen has a high molecular weight of $M_w = 340$ kDa, so it can strongly adsorb onto hydrophobic surfaces.⁴³ For PDMS and PDMS-GF, the remarkable decrease in Δf and increase in ΔD after rinsing with ASW indicate the considerable protein adsorption on the coating surface.⁴⁴ For PDMS-GP, it also has a large change of Δf and ΔD , indicating the adsorption of protein. As discussed above, this is because of the lack of PEG on the coating surface, and the PDMS with buried PEG cannot exert protein resistance. For PDMS-GF2P1, there is a decrease in protein adsorption. The protein adsorption decreases as PEG content increases, where PDMS-GF1P2 exhibits remarkable protein resistance with Δf and ΔD returning to zero after the rinsing. Note that PDMS-GF1P2 contains less PEG than PDMS-GP but shows much better protein resistance. The result clearly indicates that it is the PEG on the surface that exerts its fouling resistant ability instead of the one in bulk. In other words, PEG is enriched on the coating surface at a certain ratio of DFMA to PEG, leading to the protein resistance.

We also evaluated the antibacterial performance of the coatings for antibacterial assay after exposure to marine bacteria *Pseudomonas* sp. (Figure 8). As we know, PEG is a

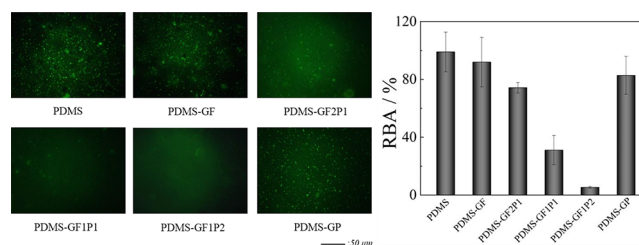


Figure 8. Fluorescence microscopy images of *Pseudomonas* sp. adhered to PDMS-GF_xPy coatings and relative bacteria adhesion (RBA).

fouling resistant polymer which can prevent the adsorption of bacteria instead of killing them.⁴⁵ A large amount of live bacteria was observed on PDMS (98% bacteria adhesion), indicating it is highly subject to bacteria adhesion. For PDMS-GF and PDMS-GP, the number of adhered bacteria was also high (more than 80% bacteria adhesion) due to the lack of fouling resistant moiety on the surface. The introduction of GF_xPy improves the antibacterial ability of PDMS-GF_xPy coatings. As PEG content increased, the adhered bacteria

decreased, and almost no bacteria were observed on PDMS-GF1P2 surface. The facts further indicate that the surface-enriched PEG improved the fouling resistant ability of the coatings.

We also evaluated the antibiofilm property of PDMS-GF α Py coatings toward *Pseudomonas* sp. biofilm (Figure 9), where

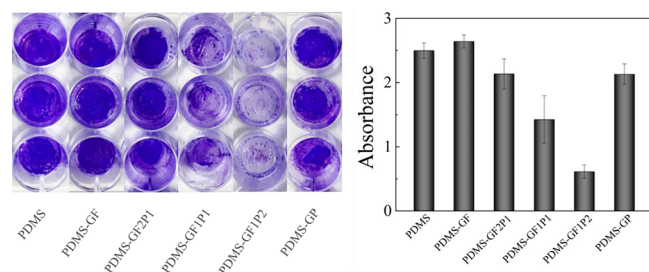


Figure 9. *Pseudomonas* sp. biofilm grown on PDMS-GF α Py coatings stained with crystal violet and the quantified value of crystal violet from the optical density test.

crystal violet was used to stain the biofilm, and the absorbance of crystal violet has a positive correlation with the biomass. PDMS was completely covered by bacterial biofilm with a high absorbance of ~ 2.5 , indicating that it cannot inhibit the growth of biofilm. PDMS-GF has an absorbance value close to that of PDMS because the surface is hydrophobic. PDMS-GP has poor antibiofilm property because the PEG concentration on the surface is not enough. When GF α Py was introduced, less biofilm was observed on the surface, and PDMS-GF1P2 showed the least biofilm with the lowest absorbance of 0.6. Note that the biofilm in some coatings did not spread on the surface of the coatings. As reported before,^{46,47} the retained biofilm becomes smaller because of the removal and redistribution of biofilm during the drying process. We added the quantified results of the biofilm coverage (Figure S5). They have similar trends with the quantified value of crystal violet from the optical density test. Clearly, the surface enrichment of PEG segments plays an important role in inhibiting biofilm formation.

Figure 10 shows the optical images of diatom *Navicula incerta* adhered on PDMS-GF α Py coatings and the diatom

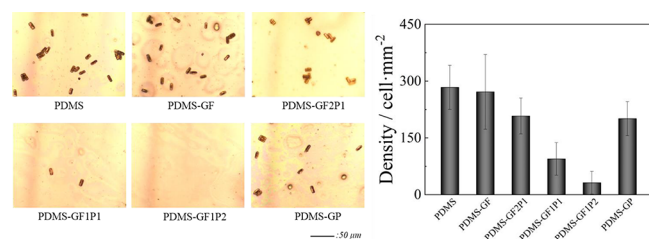


Figure 10. Optical microscopy images and density of diatom *Navicula incerta* adhered on PDMS-GF α Py coatings.

density. A large amount of diatoms (~ 280 cell·mm⁻²) adhered to the PDMS coating because diatoms favor hydrophobic surfaces.⁴⁸ PDMS-GF also had a high diatom density due to its higher hydrophobicity. PDMS-GP exhibited low antiadhesion efficiency because of low PEG content on the surface. In contrast, GF α Py improved the anti-diatom performance of the coatings where PDMS-GF1P2 has the lowest diatom density of ~ 50 cell·mm⁻². This result agrees with the protein adsorption,

bacteria adhesion, and biofilm formation assays, indicating that the amphiphile can be enriched on the coating surface under a suitable molar ratio of DFMA and PEGMA (1/2) and significantly enhanced the fouling resistance of silicone elastomer.

5. CONCLUSION

We developed a silicone elastomer with self-stratifying, nonleaching amphiphilic side chains by grafting telomers of DFMA, PEGMA, and KH590 to bis-silanol terminated silicone. The amphiphilic telomer is enriched on the surface due to the low surface energy of fluorocarbon segment and incompatibility with silicone. Such modified silicone elastomer exhibits low surface energy and low elastic modulus close to those of the unmodified one, so the coating has good fouling release performance. Moreover, it exhibits good fouling resistant performance toward bacteria adhesion, biofilm formation, and diatom adhesion under static conditions. The coating is expected to be used in marine antibiofouling.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acsapm.9b00253.

Mass loss, WCA, *Pseudomonas* sp. biofilm percentage coverage, and AFM force-depth curves of PDMS-GF α Py coatings (PDF)

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Notes

The authors declare no competing financial interest.

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